

Table 7: **Influence of the individual components of URIAL on instruct model** We report the mean MT-Bench scores for cases that have multiple different combinations, such as Rule + 1 example and Rule + 2 example. The standard deviation is always calculated on 3 runs.

In-context prompt	1st-turn	2nd-turn	Average
Empty	7.92 $\pm$ 0.08	7.18 $\pm$ 0.05	7.55 $\pm$ 0.06
Rule-only	7.80 $\pm$ 0.07	7.15 $\pm$ 0.10	7.48 $\pm$ 0.09
Rule + 1 example	7.54 $\pm$ 0.15	6.82 $\pm$ 0.34	7.18 $\pm$ 0.19
Rule + 2 examples	7.70 $\pm$ 0.17	6.84 $\pm$ 0.21	7.27 $\pm$ 0.19
URIAL	7.58 $\pm$ 0.06	6.48 $\pm$ 0.04	7.21 $\pm$ 0.06

Table 8: **Importance of question-answer matching of demonstrations for in-context alignment on instruct models.**

In-context prompt	1st-turn	2nd-turn	Average
no demonstrations	7.92	7.18	7.55
X only (no Y)	4.29	6.06	5.18
Y only (no X)	7.53	<u>6.85</u>	7.19
circular shift of Y	6.09	5.99	6.04
in-domain random Y	6.62	5.94	6.28
out-of-domain random	5.69	5.50	5.60
correct Y	<u>7.58</u>	6.48	<u>7.21</u>

B.1 INFLUENCE OF THE INDIVIDUAL COMPONENTS OF URIAL ON INSTRUCT MODEL

In Table 7, we find that different from what we observed from the base LLM, adding in-context examples, even only rules, weakens the instruction-following performance of instruct models as measured by MT-Bench performance, and more examples lead to more performance declines.

B.2 IMPORTANCE OF QUESTION-ANSWER MATCHING OF DEMONSTRATIONS FOR IN-CONTEXT ALIGNMENT ON INSTRUCT MODELS

In Table 8, we find that adding examples in the context consistently causes damage to the instruction-following performance of instruct models, even when using correct URIAL examples, which is similar to what we observe on the instruct model in Table 7. Surprisingly, it leads to a pretty bad 1st-turn score (4.29) on MT-Bench when only presenting queries and no corresponding answers in the context. After looking into the model responses, we find that the instruction model sometimes would manage to answer all queries present in the context, including the ones as demonstrations, and outputs consisting of answers to multiple questions lead to low performance on model-based automatic evaluations. This unexpected behavior is mitigated in the 2nd-turn evaluation as a complete query-answer pair is added to the context after the 1st-turn answer generation.

B.3 MANY-SHOT IN-CONTEXT ALIGNMENT ON INSTRUCT MODEL

We further test the effect of alignment with ICL on Mistral-7B-Instruct-v0.2 when increasing the number of demonstrations. We test on the same data we used in Fig. 3 and more details can be found in App. A.4. The results of many-shot in-context alignment experiments are shown in Fig. 10. We find that alignment through ICL does not further improve the instruction-following performance of instruct models, but slightly declines its capability to answer user queries.

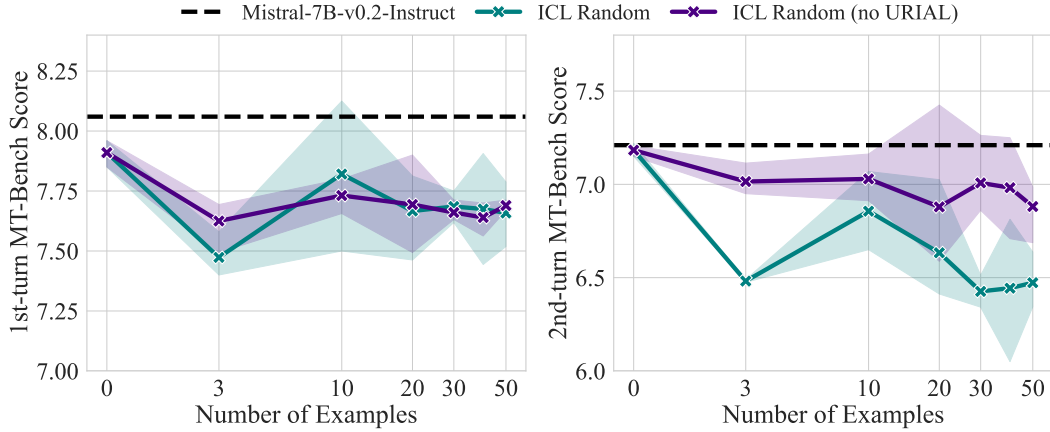


Figure 10: **Scaling the number of demonstrations for alignment with ICL on Mistral-7B-Instruct-v0.2.** We measure the alignment performance of different settings using the MT-Bench score. The base model is always Mistral-7B-Instruct-v0.2 for ICL Random and ICL Random (no URIAL).

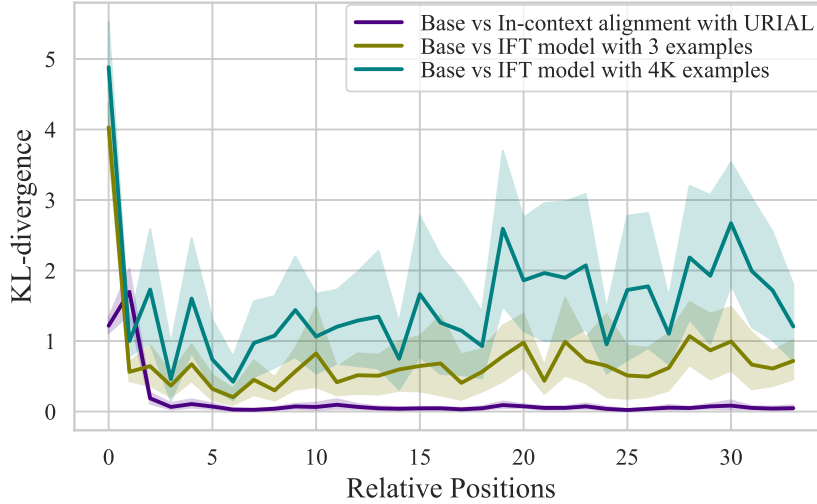


Figure 11: **KL-divergence between the base LLM and its aligned model using ICL or IFT.** The base model is always Mistral-7B-v0.2. The KL-divergence results are averaged over 25 examples.

C KL-DIVERGENCE BETWEEN THE BASE LLM AND ITS ALIGNED MODEL USING ICL OR IFT

Since we always experiment with open source LLMs with 7 ~ 8 billions parameters, we are interested in mechanistic study with logits of model outputs to reveal more insights, such as the KL-divergence between the base LLM and its aligned counterparts.

Experimental setup. We randomly select 25 query-response paired examples from SkillMix dataset as the data to calculate KL-divergence between the base LLM and the aligned models. Specifically, for each query-response paired example, two models predict the next token given the same input respectively. The model only predicts tokens from responses and each answer token after calculating KL-divergence will be added to the input for the next prediction. The response from each example is truncated to save computation costs. We always select Mistral-7B-v0.2 as the base model, and another 3 aligned models, in-context alignment using URIAL, IFT with 3 SkillMix examples, and IFT with 4k SkillMix examples, as target models to calculate KL-divergence.

Results. From Fig. 11, we observe that the probability distribution of the next token prediction, for the in-context aligned model, differs significantly only for the first few tokens, after which the

Table 9: **Transferability of greedy search IC prompt to other LLMs.** We use the 1st-turn score on MT-Bench and the length-controlled (LC) win rates on AlpacaEval (AE) 2.0 when using various IC prompts on different base models.

Model	MT-B (1st)	AE 2.0
Base model: Llama-2-7B-80k		
URIAL (3 examples)	5.19	1.81
URIAL + greedy search (1 ex.)	5.56	1.50
URIAL + greedy search (2 ex.)	5.57	1.84
URIAL + greedy search (3 ex.)	5.10	2.91
Base model: Mixtral-8x22B-v0.1-4bit		
URIAL (3 examples)	8.28	14.77
URIAL + greedy search (1 ex.)	8.36	15.74
URIAL + greedy search (2 ex.)	7.79	13.20
URIAL + greedy search (3 ex.)	8.58	14.68

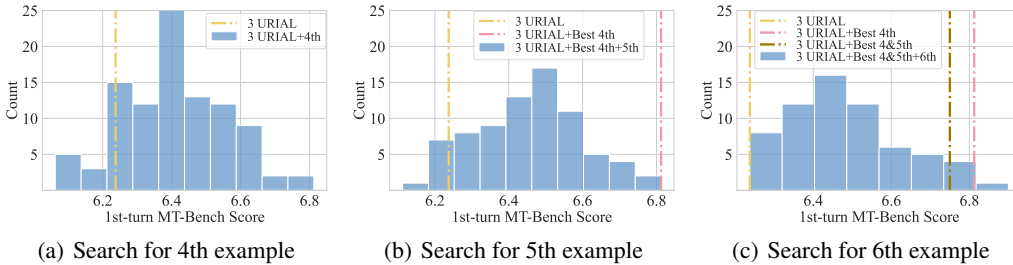


Figure 12: **The distribution of the 1st-turn MT-Bench score (GPT-4-Turbo as judge) on Mistral-7B-v0.2 obtained by adding multiple instructions from SkillMix (Kaur et al., 2024) as a 4th (a); 5th (b); 6th (c) demonstration to URIAL.** The dashed lines of various colors refer to the 1st-turn MT-Bench score of the obtained searching results. A majority of the 4th examples have positive contributions to the model’s instruction-following performance, but the improvement quickly diminishes when running the greedy search for more optimal demonstrations.

KL-divergence value rapidly drops to approximately 0. However, the KL-divergence between the IFT models and the base model does not converge to 0 and has larger KL-divergence values on average. Surprisingly, IFT on only 3 examples is already sufficient to drive the fin-tuned model far away from the base model as measured by KL-divergence compared to in-context alignment. In particular, the difference in the probability distribution of next token prediction becomes more substantial as the base model are instruction fine-tuned on more data.

D TRANSFERABILITY OF IN-CONTEXT PROMPTS

In Table 9, we report the performance of applying the in-context examples found by greedy search (added to URIAL) on Mistral-7B-v0.2 and Evol-Instruct-70k dataset (see Table. 6) to Llama-2-7B-80k (Fu et al., 2024) and Mixtral-8x22B-v0.1-4bit (Jiang et al., 2024). Adding the new examples does not provide a consistent improvement: while it can increase the MT-Bench score (+0.47 on Llama-2-7B-80k, +0.30 on Mixtral-8x22B-v0.1-4bit), it can also, in some cases, give worse results than the original URIAL. Similarly, it yields mixed results when measured by win rate on AlpacaEval 2.0, which was not optimized by the greedy search. This analysis suggests that the most effective ICL demonstrations might vary across base LLMs, potentially because of differences in their pre-training. This could further explain why URIAL underperforms on multiple models in Table 1, especially on those which have become available only recently and were not used for selecting the URIAL examples.